

Activity 4 Pre-Lab — Analysis of OTC Pain Relievers (TLC)

Organic Lab · Introduction to Chromatography

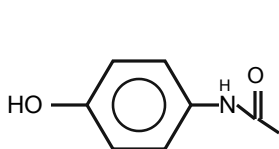
[Your name] & [partner]

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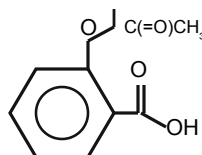
↳ This sheet tells you exactly what to write in your notebook. Copy the header onto every page, then reproduce each section below (structures & setups included).

Purpose

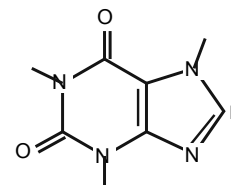
Use **thin-layer chromatography (TLC)** to separate and identify the medicinally-active compounds — **acetaminophen**, **aspirin (acetylsalicylic acid)**, and **caffeine** — in an over-the-counter pain reliever (Excedrin). Compare the R_f values of the tablet's spots against authentic standards, and examine how changing the **mobile-phase polarity** affects the separation.



acetaminophen



aspirin (acetylsalicylic acid)



caffeine (1,3,7-trimethylxanthine)

Chemical Reagents & Safety Table

Directions require a Safety & Properties table for these 5: **ethyl acetate**, **hexanes**, **acetaminophen**, **acetylsalicylic acid**, **caffeine**. SDS values below are typical Sigma-Aldrich GHS data — cross-check against the SDS tile on Labflow.

Chemical	Role in lab	MW (g/mol)	Amount used	Physical properties	Hazards (GHS)
Ethyl acetate (EtOAc)	Mobile-phase component (less-polar/moderate)	88.11	~20 mL total (in the 3 mobile phases)	Colorless liquid; bp 77 °C; density 0.902 g/mL; flash pt -4 °C; sweet odor	H225 highly flammable liquid & vapor · H319 serious eye irritation · H336 drowsiness/dizziness
Hexanes	Mobile-phase component (non-polar)	86.18	~16 mL total (in mobile phases)	Colorless liquid; bp 69 °C; density 0.66 g/mL; flash pt -22 °C	H225 flammable · H304 may be fatal if swallowed/enters airways · H315 skin irritation · H336 drowsiness · H361f suspected reproductive tox · H373 organ damage (repeat) · H411 toxic to aquatic life
Acetaminophen	Analyte / authentic standard	151.16	trace (spotting standard)	White crystalline solid; mp 168–172 °C	H302 harmful if swallowed · H315 skin irritation · H319 eye irritation · H335 respiratory irritation
Acetylsalicylic acid (aspirin)	Analyte / authentic standard	180.16	trace (spotting standard)	White crystalline solid; mp ~135 °C (decomp.)	H302 harmful if swallowed · H315 skin irritation · H319 eye irritation · H335 respiratory irritation
Caffeine	Analyte / authentic standard	194.19	trace (spotting standard)	White crystalline solid/powder; mp 234–236 °C (sublimes ~178 °C)	H302 harmful if swallowed

↳ Also used but not required in the table: **methanol** (tablet extract solvent — flammable, toxic) and a few drops of **glacial acetic acid** (added by instructor to each chamber to sharpen acidic spots).

Procedure Plan (split the page: left = your plan, right = fill in during lab)**Plan (write before lab)****Part 1 — Find the best solvent**

- Crush **2 tablets** w/ mortar & pestle → 25-mL Erlenmeyer. Add **~5 mL MeOH**, swirl 5 min, let binder settle. Label extract **"T"**.
- Prepare **3 mobile phases, 12 mL each**, in labeled 250-mL beakers (cover w/ watch glass):
 - (i) 5:1 hexanes:EtOAc
 - (ii) 1:1 hexanes:EtOAc
 - (iii) 100% EtOAc
 Instructor adds **5 drops glacial acetic acid** to each; swirl.
- Get a 4-lane TLC plate; **lightly** spot "T" on lane 1 of **3** plates (don't overload → streaking).
- Develop one plate in each chamber. Remove at solvent front, **mark the front in pencil**, air-dry.
- Visualize under **UV lamp**; pencil-circle every spot. Measure R_f ; pick the solvent giving the **best resolution**.

Part 2 — Identify the components

- New plate, best solvent, **4 lanes**: T · acetaminophen std · aspirin std · caffeine std.
- Develop, dry, UV-visualize, mark spots, measure each R_f .
- Match tablet spots to standards by R_f → identify which actives are present.

$R_f = (\text{distance spot traveled}) \div (\text{distance solvent front traveled})$. Handle plates by the edges.

Experiment Notes (in lab)

Tablet(s) used / brand: _____

Part 1 — R_f per solvent (spots low→high):

5:1 →

1:1 →

0:1 (100% EtOAc) →

Best-resolving solvent: _____

UV notes / streaking?

Part 2 — R_f match

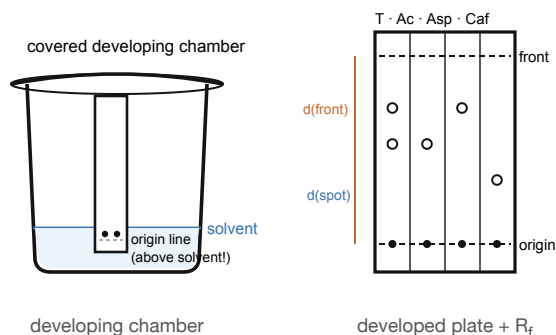
Acetaminophen std:

Aspirin std:

Caffeine std:

Tablet spots:

Actives identified in tablet:

Setup — draw these

Why acid? Aspirin is a carboxylic acid; a few drops of acetic acid in the mobile phase keep it neutral so its spot stays tight (no streaking/tailing).

Polarity idea: silica plate = polar stationary phase. More-polar mobile phase (more EtOAc) → everything moves up more (higher R_f). Pick the solvent that spreads the three spots apart the most.

Clean-up & Conclusions

- Clean-up:** pour used developing solvents into the **organic (non-halogenated) waste**; toss TLC plates, capillaries & used tablet powder in solid waste; rinse & return glassware; complete your assigned bench clean-up task.
- Conclusions (write ~4 sentences after lab):** which solvent resolved the three actives best and why (polarity); the R_f of each standard; which actives your tablet contained based on R_f matching; and one source of error (heavy spotting, non-level solvent, running the front off the plate).